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Robo-identity

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EXTENDED-ABSTRACT

Robo-Identity: Exploring Artificial Identity and Multi-Embodiment

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ABSTRACT

Interactive robots are becoming more commonplace and complex, but their identity has not yet been a key point of investigation. Identity is an overarching concept that combines traits like personality or a backstory (among other aspects) that people readily attribute to a robot to *individuate* it as a unique entity. Given people's tendency to anthropomorphize social robots, "*who* is a robot?" should be a guiding question above and beyond "what is a robot?" Hence, we open up a discussion on artificial identity through this workshop in a multi-disciplinary manner; we welcome perspectives on challenges and opportunities from fields of ethics, design, and engineering. For instance, dynamic embodiment, e.g., an agent that dynamically moves across one's smartwatch, smart speaker, and laptop, is a technical and theoretical problem, with ethical ramifications. Another consideration is whether multiple bodies may warrant multiple identities instead of an "all-in-one" identity. Who "lives" in which devices or bodies? Should their identity travel across different forms, and how can that be achieved in an ethically mindful manner? We bring together philosophical, ethical, technical, and designerly perspectives on exploring artificial identity.

CCS CONCEPTS

• **Human-centered computing → HCI theory, concepts and models**.

KEYWORDS

Artificial identity, robo-identity, embodiment, migratable AI

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Who is a robot? There are a burgeoning number of "bodies", from standard humanoid robots, e.g., Nao, to more creative renditions, e.g., a synchronized trio of robots as one[1] or a more abstract Jibo[2]. Yet thus far, *who* resides in artificial bodies and whether their identities can fluidly travel across different technologies remains as a fascinating, but under-explored research area. As a topic, *identity* has not been an explicit focus of research in Human-Robot Interaction (HRI), and in neighboring fields of robo-philosophy, HCI, and design; whose identity is housed where and why deserves to be more thoroughly investigated. Hence, this half day workshop aims to collaboratively discuss the problem of artificial identity or robo-identity that is brought forth by increasingly diverse interfaces and embodiments, in a multi-disciplinary manner. We summarize how interrelated philosophical, designerly, and engineering perspectives on identity are relevant to HRI below.

1 PHILOSOPHY

Discussions on *human* identity is abundant in philosophy, e.g., works by Locke [4] or Parfit [8] among others, but philosophical views are yet to be fully integrated into thinking about robo-identity. One line of thought that is relevant in HRI would be the distinction between substance vs. consciousness view on identity, i.e., am I the same "me" based on the continuance of the body that I have vs. continued memory? Even if a person loses their memory over time, there is a semblance of a singular identity that we often attribute to them based on character traits or psychological unity. When one's human identity is psychologically distributed across different voices, it is normally seen as a psychological disorder, e.g., schizophrenia or dissociative identity disorder [12].

But unlike our human identity that we experientially equate with a body or psychologically normative stream of memories, an artificial identity can more easily travel across various digital forms, e.g., all-in-one "Siri" across multiple devices. When one's smartwatch and smart speaker have a unified identity within an ecosystem, or

[1] https://www.wired.co.uk/article/panasonics-aug-lab-wants-to-use-robotics-to-enrich-peoples-lives
[2] https://jibo.com/

when various apps on a phone have different identities, we would not think of this as "broken" or "schizophrenic" technology. Thus, our expected norms about human vs. robot identity may be different. For robo-identity, we do not yet know if a single identity across various devices and modalities is a desirable goal, or what it would even mean to construct a unified artificial identity as a sum of which disparate parts, e.g., hardware, software, and data. Furthermore, ethical issues are many, ranging from "hijacked" robo-identity during body transference to fragmented user experience and autonomy. A related concern is whether a robot's identity can and should be treated in a similar way as a person's identity. Lastly, given prior research on how technology can influence one's moral sense of self through, e.g., compassion [3], there is potential in exploring how to best design artificial identity to help with our own identity development.

2 DESIGN

Previous work in the field of HRI has looked into the notion "migrating robots" [1, 11], and found that participants could easily perceive when one "identity" moved from one device to another [7]. However, it was not clear when would it be beneficial to do so. More recent work has taken a design research approach to explore how migrating agents are perceived, specifically on when they can provide value for a range of contexts, e.g., in personal or professional settings [6], or in service contexts [10]. Initial findings show that people may favor migrating robots across a single service, or in stressful situations. However, many design questions on migrating agents are still left unanswered—what new design opportunities do migrating agents offer? In which situations do individuals prefer agents that do not migrate? How are migrating robot identities perceived across a range of contexts? In the workshop we intend to discuss the work that exists in this complex design space, and identify some of the open-ended questions that require further investigation.

3 EMBODIMENT AND INFORMATION

Who robots come across as has consequences on our actions. Robots' various forms of embodiment are known to change our behavior and situational awareness [2, 5]. The embodiment type, e.g., a VR agent vs. a physical robot, impacts people's perceived enjoyment of interacting with a robot [9]. Hence, how to sync identity traits across different bodies is an engineering and design challenge.

Humans interact through various means of communication, from computer-mediated communication to face-to-face interaction, with a singular identity, yet robot conversational partners' identities in different contexts remain unclear. This is due to physical constraints that arise with robot embodiment and related multimodal behavior. The identity features of the migratable agent such as voice and visual characteristics are restricted based on the software and hardware capabilities of the embodiment. This leads to the challenge of how to map an agent's identity features to its embodiment and multimodal capabilities during its identity migration.

Another issue is not only an agent's identity, but how much knowledge it has on users' identity for each type of body it moves to. A migratable agent lacks contextual understanding of users' information. This opens up the question of how might an agent behave when it transfers to another form. For instance, an interesting challenge is on how much user information an agent should take with it when it migrates to a different embodiment in public vs. private settings. This workshop aims to discuss the design and engineering challenges over how should robots migrating through different forms of embodiment use the available channels of communication to convey their identity to users.

4 ORGANIZERS

Minha Lee is an assistant professor at the Eindhoven University of Technology at the department of Industrial Design, with a background in philosophy, digital arts, and HCI. Her research is about morally relevant interactions with technological agents like robots or chatbots. Her recent work explores how we can explore our moral self-identity through conversations with digital entities, e.g., via acting compassionately towards a chatbot.

Dimosthenis Kontogiorgos is a PhD student at KTH Royal Institute of Technology and is interested in how conversational agents' embodiment and non-verbal behaviors affect the process of establishing, maintaining and repairing common ground. He has previously co-organized the Young Researchers' workshop in dialogue systems (YRRSDS 2019) and the doctoral consortium of Interspeech 2017.

Ilaria Torre is a postdoctoral researcher at KTH Royal Institute of Technology. She is interested in artificial agents' communicative and affective cues, particularly through speech, and how these affect cooperation and trust in Human-Agent Interaction. She previously worked as a postdoc at Trinity College Dublin, and holds a PhD from the University of Plymouth. She organized a satellite workshop of HAI 2018, "Measuring and Designing Trust in Human-Agent Interaction", the UK Speech 2018 conference, and the ColLaboratoire 2016 Summer School.

Michal Luria is a Ph.D. candidate in the Human-Computer Interaction Institute at Carnegie Mellon University. Her work uses Research-through-Design and Speculative Design approaches to investigate how intelligent personal agents and social robots can and should behave and interact. Michal's recent work has looked into why, and in which contexts, it would be beneficial for agents to 're-embody' as part of their interaction with people. She had previously co-organized the 2019 HRI Pioneers Workshop.

Ravi Tejwani is a graduate student in Personal Robots Group at Massachusetts Institute of Technology. His work focuses on modeling the behavior of the conversational AI agent with the user from verbal and non-verbal cues using deep learning techniques. Ravi's most recent work on Migratable AI explored the migration of AI agents across different robotic embodiments.

Matthew J. Dennis is a Marie Skłodowska-Curie Research Fellow in the Department of Values, Technology, and Innovation at TU Delft. He specializes in digital well-being, focusing on how technology can increase human flourishing. He writes on how we can improve the design of emerging technologies (social robots, virtual assistants, self-care apps, etc.), as well as how digital well-being is affected by gender, education, and economic factors. Questions relating to identity feature throughout his work, most recently in his article on how social robots can be used to facilitate processes

of personal transformation.

André Pereira is a researcher at KTH Royal Institute of Technology. He completed his PhD at IST Technical University of Lisbon, worked as a Postdoc at Yale University, and as a Senior Research Associate at Disney Research. André's primary research goal is to create autonomous embodied agents, typically robots, that can socially interact with humans throughout extended periods. Concerning the topics of this workshop, he has worked on agents that can 'migrate' data and visual identity between social robots and mobile phones. He is now interested in how agents that can migrate between devices can affect users' motivation to interact with digital assistants.

5 CONFIRMED SPEAKER

Iolanda Leite is an Assistant Professor at the School of Electrical Engineering and Computer Science at the Royal Institute of Technology, KTH. Her research bridges together many relevant concepts to identity, such as how to design for engagement, play, and affective interaction while considering different types of embodiment. She holds a PhD in Information Systems and Computer Engineering from IST, University of Lisbon. She had postdoctoral appointments at Yale University and at Disney Research. Iolanda's research interests are in the areas of AI and human-robot interaction. She is the co-editor in chief of AI Matters, the newsletter of the ACM Special Interest Group in Artificial Intelligence (SIGAI).

6 SCHEDULE OF ACTIVITIES PLANNED

During the workshop, we plan to explore divergent and convergent themes under the heading of robo-identity. Activities are below.

- 1:30–1:45 Introduction and agenda
- 1:50–3:00 Presentations of papers (5-6 minutes each)
- 3:10–3:20 Invited speaker
- 3:20–3:25 Research synthesis 1: breakout rooms
- 3:30-4:00 Research synthesis 2: breakout rooms
- 4:05-4:35 Research synthesis 3: breakout rooms
- 4:40-5:10 Reconvene for a sharing session
- 5:10-5:30 Future steps and closing

Our goal is to bridge gaps between relevant disciplines in order for identity as a topic to be treated in multi-faceted ways. Since we expect early ideas on identity rather than fully developed works, we will ask authors to present in PechaKucha style[3], i.e., 15 to 20 slides with a maximum of 20 seconds per slide (though this is a recommendation rather than a strict requirement). The hope is in sharing authors' vision on identity that can inspire co-attendees in an easily accessible manner. We will keep presentations shorter and *synthesis* oriented discussions longer with breakout rooms; participants will be distributed to breakout rooms with people from different backgrounds. By building on short, vision-oriented presentations, discussion sessions would provide ample time for brainstorming research questions and agendas. A sharing session that follows would identify key areas people want to work together on for future research.

[3]https://www.pechakucha.com/

7 WORKSHOP PLANNING AND OUTCOMES

Based on organizers' communities, we hope to include researchers in HRI, design research, philosophy, conversational user interfaces, and HCI fields. By doing so, we will, for example, bring the RoboPhilosphy and HRI communities into a discussion regarding overlapping interests on robo-identity. All papers will be reviewed by our program committee. An overview of the workshop (including our program committee) can be found on the website[4]. Our website will serve as a repository for exploratory papers of participants. Accepted papers will also be posted on arXiv, Hence, we are mobilizing for a broader awareness of robo-identity as a topic that may concern many groups of researchers. At the end of our workshop, we will have shared research questions and guiding thoughts as starting seeds of collaboration, e.g., for more elaborate position papers with co-authors from different disciplines. Currently, many researchers in RoboPhilosophy, HRI, design, and engineering implicitly deal with the problem of identity and multi-embodiment, and our workshop will bring the topic of identity to a shared foreground.

8 ACKNOWLEDGEMENTS

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